

Progress Report

Situation-Aware Player Similarity via Trait–State Decomposition

Seo Won Yi

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1 Introduction and Problem Formulation

Task. Given a query soccer player p , the objective is to retrieve a ranked list of similar players who function as true *substitutes*—meaning they exhibit *similar on-ball decision-making* when subjected to the *exact same game constraints*. This approach deliberately abandons surface-level aggregate statistics (e.g., total goals, pass completion rates) in favor of capturing fundamental behavioral and tactical tendencies.

Formalizing Similarity. Soccer decisions are inherently *contextual*; a “pass from (x, y) ” is not a uniform behavior, as it is heavily dictated by passing lanes, defensive proximity, and teammate availability. Therefore, any robust similarity metric must strictly condition on the game state.

Let s denote the **masked pre-action state**—including spatial freeze-frame context and all contextual metadata available prior to the decision— y the **action** taken (e.g., action type, direction, and length attributes), and z_p a latent **player trait** embedding capturing individual style. I define players p and q to be similar if their **conditional action distributions** converge across realistic game states:

$$p \sim q \iff \mathbb{E}_{s \sim \mathcal{S}} \left[d(P_\theta(y \mid s, z_p), P_\theta(y \mid s, z_q)) \right] \text{ is small,}$$

where P_θ denotes the learned conditional model, $d(\cdot, \cdot)$ is a divergence metric (e.g., Jensen–Shannon), and $\mathcal{S}$ is the empirical distribution of plausible in-game states. The operational goal is to learn a global representation z_p that captures persistent preferences (e.g., progressive risk appetite) that remain after fully controlling for the tactical environment.

2 Data and State Representation

The system utilizes StatsBomb 360 data, where each decision point is an **on-ball event** containing actor identity, ball location, contextual metadata, and crucially, a **360 freeze-frame snapshot** providing the spatial coordinates of visible players at that instant.

The freeze-frame is what makes the state s well-defined: it encodes the spatial constraints and option sets. Without it, the model cannot mathematically distinguish a creative through-ball executed under heavy pressure from a routine square pass in open space. Furthermore, because tactical decisions are path-dependent, events are grouped into **possessions**. A possession k is formulated as a temporal sequence:

$$\mathcal{P}_k = \{(s_{k,1}, y_{k,1}, p_{k,1}), \dots, (s_{k,T_k}, y_{k,T_k}, p_{k,T_k})\},$$

ensuring every analyzed action possesses a coherent tactical history.

3 Methodology: Trait–State Architecture

The core methodology is a trait–state model trained via masked imitation, designed to summarize “what the situation affords” (state) and how a specific player modulates decisions within it (trait).

3.1 Preventing Label Leakage via Strict Masking

A critical risk in imitation learning is label leakage; if inputs contain fields that reveal the action, the model will trivially copy the target without learning the underlying decision structure. I enforce a strict pre-action state by masking action types, end locations, displacements, and outcomes.

Crucially, the actor’s **positional role** is strictly masked from the event features fed into the state encoder. This prevents the Graph Neural Network from relying on broad positional heuristics (e.g., “Center Backs usually clear the ball”) to build the situation representation, forcing it to genuinely understand the spatial and relational geometry of the play. Consequently, the behavioral variance is forced to flow through the player-node pathway and into the learned trait embedding z_p .

3.2 Structured State Representation via Heterogeneous Graphs

A possession is a multi-agent interaction, not merely a flat sequence. To inject a relational inductive bias, I model each possession as a heterogeneous graph G_k . This structure jointly reasons over *temporal evolution* (event-to-event edges with continuous time-deltas), *actor dynamics* (player-to-event edges), and *spatial constraints* (context edges connecting off-ball players to focal events via geometric offsets).

As illustrated in Figure 1, a 2-layer Heterogeneous Graph Attention Network (GATv2) propagates these relations, yielding event-level encodings $h_{k,t}^{\text{event}}$ and player-in-context encodings $h_{k,t}^{\text{player}}$ in a shared latent space:

$$h_{k,t}^{\text{event}}, h_{k,t}^{\text{player}} = f_{\theta}(G_k, \tilde{x}_{k,1:T_k}) \in \mathbb{R}^d.$$

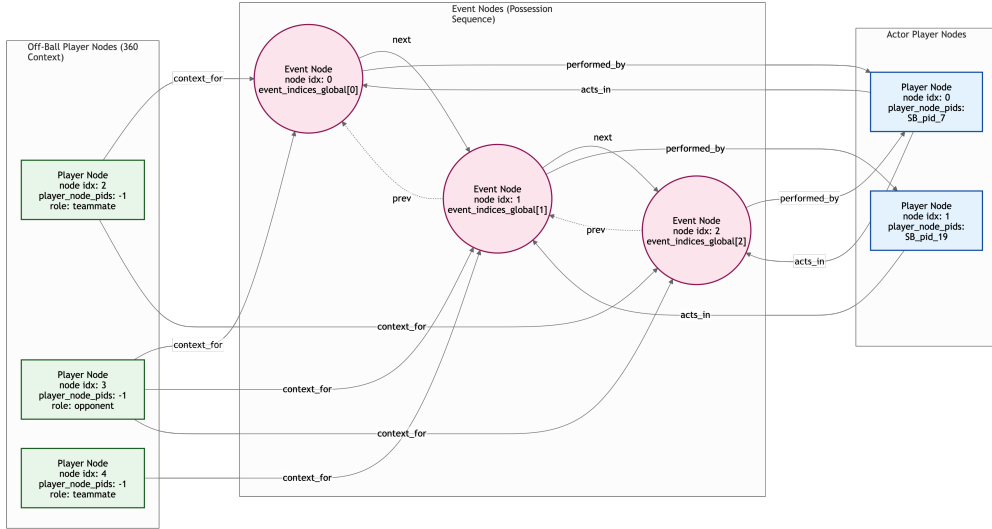


Figure 1: Heterogeneous graph architecture mapping temporal, actor, and spatial context relations within a possession.

3.3 Trait Extraction and Multiplicative Conditioning (FiLM)

To extract a global trait from highly variable contexts, I aggregate per-possession player-in-context embeddings via end-to-end learned attention pooling: $z_p = \sum_i \alpha_i h_{\text{player},i}$. This allows the model to up-weight high-leverage situations that uniquely reveal style, de-emphasizing routine recycling.

To enforce conditional dependence, I apply Feature-wise Linear Modulation (FiLM). Simple concatenation permits the network to down-weight or ignore latent trait dimensions; FiLM instead

enforces multiplicative dependence, making trait variation load-bearing for prediction.

While position is hidden from the state encoder, I utilize a **dual-channel position design** at the prediction head, concatenating z_p with a separate positional embedding to formulate the conditioning vector (see Figure 2). By reintroducing nominal position at this final stage, the model uses it as a structural baseline. This forces z_p to encode fine-grained, *intra-positional* stylistic deviations (e.g., how a specific player differs from the baseline of an average center-back) rather than wasting representational capacity on macro-tactical roles:

$$\text{FiLM}(h_{k,t}; z_p, \text{pos}) = (1 + \gamma([z_p; \text{pos}])) \odot h_{k,t} + \beta([z_p; \text{pos}]).$$

This mechanism predicts the action, ensuring that z_p must mathematically explain individual behavioral variations within a shared positional role.

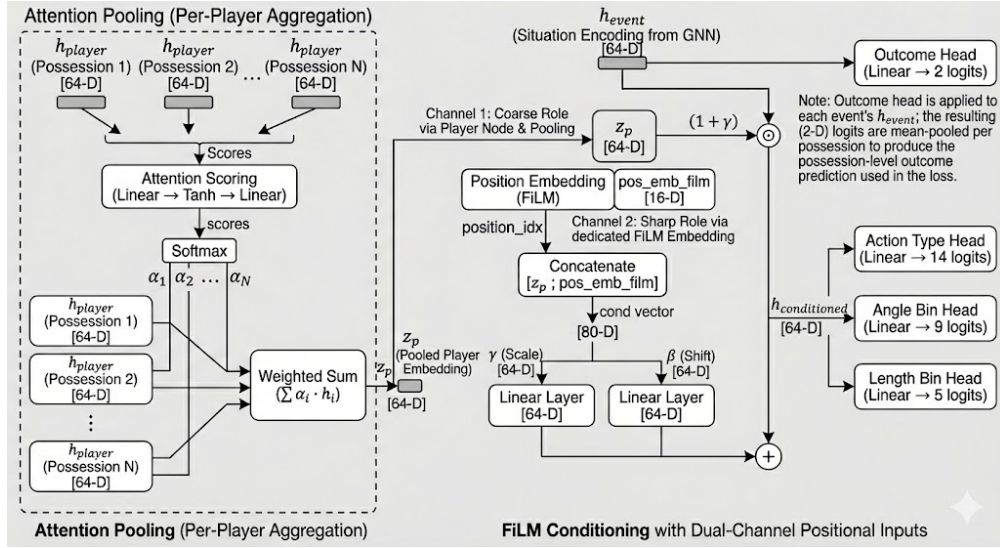


Figure 2: Neural network architecture detailing attention pooling and FiLM conditioning with dual-channel positional inputs.

4 Training Objectives and Conceptual Validation

The model is optimized via a combined loss function aligning with the theoretical definition of similarity:

1. **Masked Imitation:** A Focal Loss optimizes action prediction $P_\theta(y | h, z_p)$, handling heavy class imbalances (e.g., rare shots vs. frequent passes).
2. **Contrastive Regularization (InfoNCE):** Intra-player consistency and inter-player distinctiveness are enforced using *same-position-group hard negatives*, forcing the model to learn fine-grained individual differences rather than coarse tactical clusters.
3. **Uniformity on z_p :** A Gaussian-potential uniformity loss applied directly to pooled trait embeddings z_p explicitly pushes representations apart on the unit hypersphere, preventing embedding collapse and widening cosine similarity gaps.
4. **Value/Outcome:** An auxiliary Binary Cross-Entropy loss predicts possession outcomes, injecting a task-relevant inductive bias.

5 Results

5.1 Experimental Setup and Held-Out Prediction

To validate the underlying behavioral model before analyzing player similarity, predictive capabilities were evaluated on a held-out test set. The dataset relies on StatsBomb 360 freeze-frame data, comprising approximately 51,000 possessions and 737,000 on-ball events. To strictly prevent label leakage, data was split by *match identifier* (70% train, 15% validation, 15% test). Consequently, the model never observes any events from a test match during training.

The model predicts discrete targets for action type (14 classes), angle bin (9 directional sectors), and length bin (5 displacement magnitudes), alongside binary outcome predictions for `ends_in_shot` and `ends_in_goal`. Because baseline actions such as Passes ($\sim 38\%$) and Carries ($\sim 33\%$) heavily dominate the dataset, Macro F1 scores are prioritized to ensure rare but impactful actions (e.g., Shots, Dribbles) are accurately modeled. Table 1 and Figure 3 detail these results.

Table 1: Test-set evaluation metrics for action and outcome prediction heads. Values validate the model’s ability to reconstruct behavior from masked game states.

Task	Metric	Value
Action Type (14-way)	Accuracy	0.7831
	Macro F1	0.6591
Angle Bin (9-way)	Accuracy	0.7078
	Macro F1	0.6530
Length Bin (5-way)	Accuracy	0.7503
	Macro F1	0.6819
Outcome: Shot	Accuracy	0.9461
	AUC-ROC	0.9704
Outcome: Goal	Accuracy	0.9882
	AUC-ROC	0.9872

These metrics confirm that the network successfully parses the masked game state to reconstruct observed actions, establishing a reliable foundation for extracting individual player traits.

5.2 Embedding Space and Global Similarity

Global player embeddings $z_p \in \mathbb{R}^{64}$ were generated by aggregating each player’s historical possessions via the learned attention-pooling module. Players with fewer than 50 recorded possessions were excluded to maintain representation stability. Similarity between a query player q and candidate players was subsequently computed using pairwise cosine similarity across the Z matrix.

As shown in Figure 4, a global Principal Component Analysis (PCA) projection of the embedding space demonstrates clear structural clustering by primary position subgroup (Goalkeeper, Center Back, Full Back, Defensive Mid, Central Mid, Wide Mid, Forward). Because explicit positional labels were masked during the event-encoding phase, this spatial separation indicates that the model successfully recovers macro-tactical roles purely from relational behavior and spatial context.

5.3 Situation-Level Behavioral Alignment

To validate the latent similarity, I perform situation-level counterfactual analysis. By holding the situational state h constant and substituting different candidate traits z_q , I compare the predicted

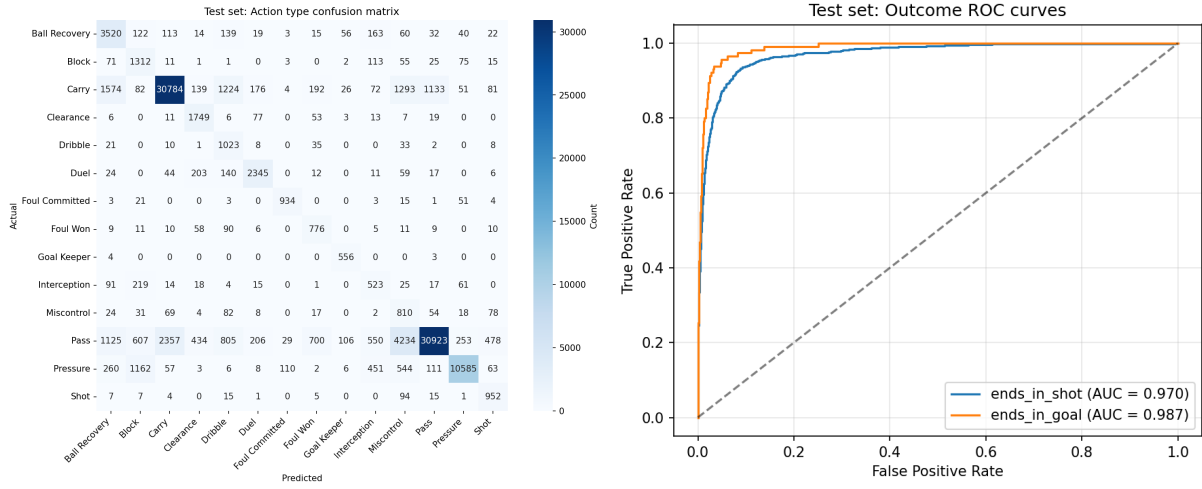


Figure 3: Left: Confusion matrix for the 14-class action type prediction. Right: ROC curves for possession-level outcome predictions.

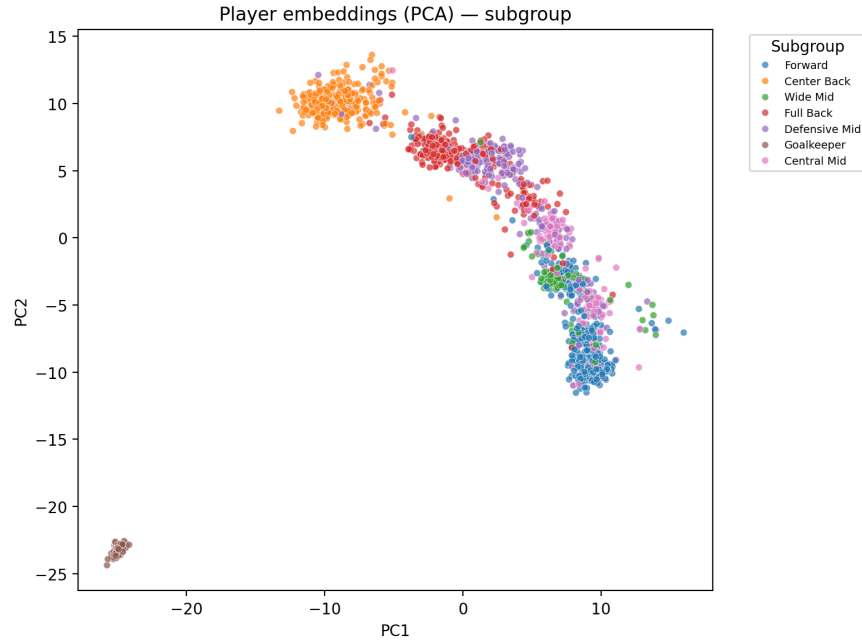


Figure 4: A 2D PCA projection of the global player embedding space, colored by coarse position group. Players with similar tactical profiles naturally cluster together.

action distributions. While limited by the "ghosting" fallacy (swapping a player physically alters preceding gravitational pull on defenders), this strictly satisfies the formal definition of conditional similarity. Despite this physical inconsistency, the counterfactual evaluates the learned conditional policy under controlled state equivalence, providing a principled operational test of trait alignment.

For a given query player, up to $S = 5$ real game events were sampled. The local situation vector $h_{\text{event}} \in \mathbb{R}^{64}$ was computed and then frozen. We then injected the global latent traits z_p of both the query player and their top- k nearest neighbors through the FiLM conditioning layer to generate conditional action distributions $\hat{\pi}_p(\cdot | s)$.

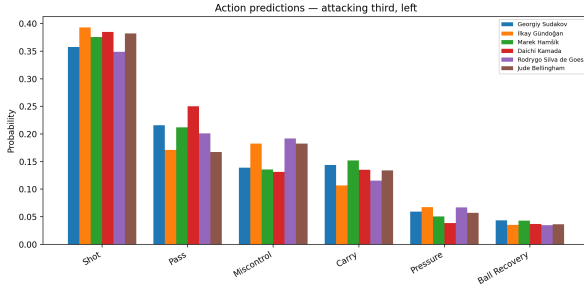


Figure 5: Probability distributions of top action types for query player and neighbors.

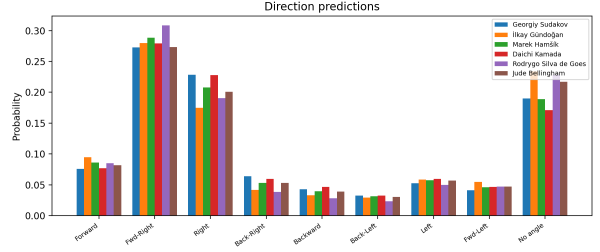


Figure 6: Directional probability distributions for query player and neighbors.

Qualitative reports and distribution comparisons (Figures 5 and 6) provide strong evidence for behavioral alignment. When subjected to the exact same spatial constraints and temporal history, nearest neighbors in the z_p space output highly correlated probability distributions for action type, angle, and length. Conversely, significant divergence in $\hat{\pi}_p$ is observed when traits of low-similarity players are substituted, confirming that the trait embeddings capture fine-grained individual decision-making policies rather than generic contextual responses.

6 Future Implementations and Limitations

While the current trait-state architecture provides a robust framework for situation-aware player similarity, several theoretical and practical avenues remain for future refinement. This section outlines critical vulnerabilities identified in the current design and proposes concrete methodological improvements.

6.1 Refining Graph Relational Biases

Currently, the system utilizes a single `context_for` edge type to map 360-degree spatial context. Consequently, the GNN relies heavily on node-level features (e.g., `is_possession_team`) to distinguish between a teammate providing a passing lane and an opponent applying pressure. Future iterations will split this into distinct (`teammate`, `context_for`, `event`) and (`opponent`, `context_for`, `event`) edge types. This structural change will allow the HeteroConv layers to learn dedicated projection matrices and attention weights for offensive versus defensive spatial constraints.

6.2 Counterfactual-Safe Situation Encoding

For the trait-swapping evaluation to be mathematically sound, the situation encoding h_{event} must be strictly player-agnostic. Currently, message passing includes (`player`, `acts_in`, `event`) edges, creating a risk that h_{event} inadvertently absorbs the original actor’s identity or role profile before the candidate trait z_q is evaluated. Implementing a counterfactual-safe encoder—such as applying a

stop-gradient or zero-masking the actor’s features on the `acts_in` pathway during extraction—will ensure h_{event} is conditioned solely on environmental context and temporal history.

6.3 Continuous Action Spaces

The current architecture discretizes continuous action attributes (angles and displacement lengths) into predefined bins. This approach can artificially penalize near-boundary predictions (e.g., treating a 15m and 16m pass as entirely separate classes). Transitioning the output heads to a Mixture Density Network (MDN) or utilizing direct regression (e.g., Huber loss) for pass and carry vectors (dx, dy) would yield a more continuous, high-fidelity probabilistic model of spatial decision-making.

6.4 Substitute-Specific Evaluation Metrics

The existing evaluation relies on action and outcome prediction accuracy, serving as a proxy rather than a direct measure of similarity. Future development will introduce metrics that specifically target substitute quality:

- **Policy Distance:** Measuring the Jensen–Shannon (JS) divergence between the predicted action policies $\pi_p(\cdot \mid s)$ and $\pi_q(\cdot \mid s)$ over a canonical set of standardized situations to ensure that cosine similarity in the z_p space mathematically aligns with identical decision-making.
- **Human-Annotated Benchmarks:** Developing a reference dataset of known similar players (derived from expert scouting labels or established tactical clusters) to compute rank correlation (e.g., Hit@K). This will ground the latent embedding space in expert domain intuition.